in which it travels. Velocity of light is more in a rarer medium (low optical density) than in a denser medium (high optical density).

6.7.1 Refraction of light from a plane transparent surface

When a ray of light travels from optically rarer medium to optically denser medium, it bends towards the normal. (Figure 6.11a).

When a ray of light travels from an optically denser medium to an optically rarer medium it bends away from the normal. (Figure 6.11b).

A ray of light incident normally on a denser medium, goes without any deviation. (Figure 6.11c).

6.7.2 The laws of refraction of light

Laws of refraction, also known as Snell's law of refraction are given below as:

- The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a light of given colour and for the given pair of media.

If i is the angle of incidence and r is the angle of refraction, then

$$\frac{\sin i}{\sin r} = \text{constant}$$

This constant is called the refractive index of the second medium with respect to the first medium. It is generally represented by the Greek letter, ${}_1\mu_2$ (mew).

Note: The refractive index has no unit as it is the ratio of two similar quantities.

The refractive index of a medium is also defined in terms of speed of light in different media.

$$\mu = \frac{\text{Speed of light in vacuum or air } (c)}{\text{Speed of light in the medium } (v)}$$

$$\text{In general, } {}_1\mu_2 = \frac{\text{Speed of light in medium 1}}{\text{Speed of light in medium 2}}$$

Problem 4

The speed of light in air is 3×10^8 ms^{-1} and in glass it is 2×10^8 ms^{-1}. What is the refractive index of glass?

Solution:

$${}_a\mu_g = \frac{3\times10^8}{2\times10^8} = \frac{3}{2} = 1.5$$

Problem 5

Light travels from a rarer medium to a denser medium. The angles of incidence and refraction are respectively 45° and 30°. Calculate the refractive index of the second medium with respect to the first medium.

Solution:

$${}_1\mu_2 = \frac{\sin i}{\sin r} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2} = 1.414$$

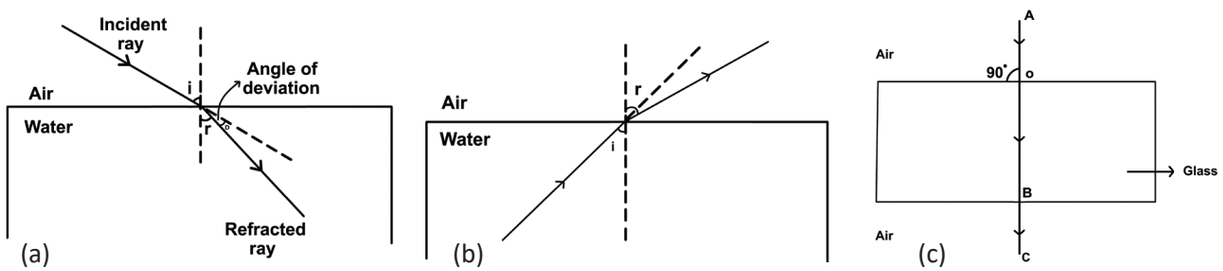


Figure 6.11 Refraction of light from a plane transparent surface

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